

Topological Anomaly Score and Persistent Entropy for Early Detection of Chaos Transition in Double Pendulum Dynamics

— Revised Version: Incorporating Reviewer Recommendations —
A Topological Langevin Dynamics (TLD) Framework

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Abstract

Chaos detection traditionally relies on Lyapunov exponents (LLE), which become significant only after instability has developed. This paper introduces a topology-based early-warning framework — the Topological Langevin Dynamics (TLD) framework — combining Persistent Homology, Persistent Entropy (H_p), and the novel Topological Anomaly Score (TAS). TAS measures the standardized maximum barcode lifetime relative to the baseline distribution, providing a sensitive indicator of topological structural changes in phase space.

We establish three formal theorems connecting (T1) Jacobian spectral growth to persistent lifetime bounds [proof sketch; rigorous proof in progress], (T2) Persistent Entropy as a verified early-warning signal satisfying $t_p^T < t_{le}$ by 1.8 seconds, and (T3) anisotropic metric sensitivity to dynamically critical directions. A revised understanding of TAS is presented: TAS detects sustained topological anomalies rather than initial chaos onset.

Numerical experiments on the double pendulum across three dynamical regimes (periodic, quasi-periodic, chaotic) validate these findings. An adaptive α -tuning algorithm for the anisotropic metric is introduced to enhance robustness. The framework extends naturally to power systems, climate tipping points, and aerospace control.

Keywords: *persistent homology; Topological Anomaly Score; Persistent Entropy; double pendulum; chaos detection; Langevin dynamics; early warning; anisotropic metric*

1. Introduction

The double pendulum is the canonical model of deterministic chaos: its governing equations are fully deterministic, yet arbitrarily small perturbations in initial conditions produce exponentially diverging trajectories. Traditional chaos indicators — Lyapunov exponents, Poincaré sections, fractal dimension — are fundamentally local: they measure the rate at which nearby trajectories diverge, but typically become significant only after the chaotic regime has fully developed.

This paper proposes a complementary, global topological approach. Rather than tracking trajectory divergence, we track the structural evolution of phase-space point clouds via persistent homology. The central objects are the Persistent Entropy H_p and the novel Topological Anomaly Score (TAS) — both derived from barcode lifetimes of the Vietoris-Rips filtration applied to the state-space trajectory under an anisotropic metric that emphasizes velocity-sensitive directions.

This framework is a direct extension of our prior work. In [A], we showed that the Eyring-Kramers spectral gap $\lambda_1(\beta) \sim C \cdot \exp(-\beta \cdot H^*)$ governs the mixing time of Langevin MCMC in nonconvex energy landscapes. In [C], we showed that the fragmentation index $F(\epsilon)$ of near-optimal sublevel sets correlates strongly with

computational hardness ($r = -0.894$). The present paper applies the same philosophy to continuous dynamical systems: we ask whether topological complexity of the phase-space cloud predicts the onset of chaos.

This revised version incorporates three major improvements responding to critical review: (1) Theorem T2 is corrected to reflect the empirically validated ordering $t_p^T < t_{le}$, with TAS repositioned as a sustained-anomaly detector; (2) the proof status of T1 is clarified as a preliminary sketch pending rigorous completion; (3) an adaptive α -tuning algorithm is introduced in Section 2.3 to automate the anisotropic metric calibration.

1.1 Main Contributions

The main contributions of this paper are:

- (C1) Formal framework: anisotropic metric, persistent homology, H_p and TAS definitions.
- (C2) Three theorems with corrected T2: the empirically confirmed early-warning ordering $t_p^T < t_{le}$ (1.8s lead time), and TAS as sustained-anomaly indicator.
- (C3) Adaptive α -tuning algorithm (Section 2.3): gradient-free Bayesian optimization of the anisotropic metric weight.
- (C4) Numerical validation across three dynamical regimes with sliding-window early-warning analysis.
- (C5) Connection to the TLD research programme [A, B, C] and applications to power systems, climate, and aerospace.

2. State-Space Representation and Anisotropic Metric

2.1 State-Space Point Cloud

The double pendulum state vector is $X(t) = (\theta_1, \omega_1, \theta_2, \omega_2) \in \mathbb{R}^4$. A trajectory of duration T , sampled at interval Δt , generates the point cloud:

$$P = \{X(t_i)\}_{i=1}^T, \quad N = T / \Delta t.$$

The topological structure of P encodes the phase-space geometry. In the periodic regime, P forms closed curves (tori). In the chaotic regime, P fills a complex fractal attractor. We make this distinction quantitative via persistent homology.

2.2 Fixed Anisotropic Metric

Standard Euclidean distance weights angle and velocity components equally. However, chaos manifests first through velocity divergence. We define the anisotropic metric:

$$da(i,j) = \sqrt{[\alpha^2(\omega_{1,i} - \omega_{1,j})^2 + \alpha^2(\omega_{2,i} - \omega_{2,j})^2 + (\theta_{1,i} - \theta_{1,j})^2 + (\theta_{2,i} - \theta_{2,j})^2]}$$

where $\alpha > 1$ controls sensitivity to velocity fluctuations. In Section 2.3, we introduce an adaptive algorithm to optimize α automatically. For the baseline experiments reported in Section 5, we use $\alpha = 2.0$ (fixed).

Theorem T3 — Anisotropic Sensitivity

Let X^c be a chaotic trajectory and X_p a periodic trajectory with the same initial energy. For $\alpha > 1$, the maximum H_0 barcode lifetime $L_{\max}$ under d_α satisfies:

$$L_{\max}(X^c; d\alpha) / L_{\max}(X_p; d\alpha) > L_{\max}(X^c; d_1) / L_{\max}(X_p; d_1).$$

That is, the anisotropic metric amplifies the topological distinction between chaotic and periodic trajectories. The ratio grows as $O(\alpha^2)$ in the velocity-dominated regime.

2.3 Adaptive α -Tuning Algorithm (NEW)

A fixed value $\alpha = 2.0$ may not be optimal across all dynamical systems or parameter regimes. We introduce an adaptive algorithm that automatically selects α^* by maximizing discriminability between dynamical states while minimizing sensitivity to noise.

2.3.1 Objective Function

The objective trades off two criteria:

Discriminability Maximization: maximize the separation of H_p (or TAS) values between identified periodic and chaotic windows.

Stability Optimization: minimize the coefficient of variation $CV(H_p) = \sigma(H_p)/\mu(H_p)$ within a noise-corrupted trajectory segment, ensuring the early-warning signal remains consistent.

2.3.2 Gradient-Free Bayesian Optimization

Because the mapping $\alpha \rightarrow H_p(\alpha)$ is non-differentiable through the Vietoris-Rips filtration, we adopt a gradient-free approach. The algorithm proceeds as follows:

Step 1 (Initialization): Evaluate H_p and TAS for a small set of candidate values $\alpha \in \{1.0, 1.5, 2.0, 3.0, 5.0\}$ on a short trajectory segment (first 10% of T).

Step 2 (Gaussian Process Prior): Fit a Gaussian Process (GP) model $\mu^{\text{asp}}(\alpha)$, $\sigma^{\text{asp}}(\alpha)$ to the observed discriminability scores, treating α as a 1D input.

Step 3 (Acquisition): Select the next candidate via Expected Improvement: $\alpha_n^{\text{ex}_t} = \text{argmax} [EI(\alpha)] = \text{argmax} [(\mu^{\text{asp}}(\alpha) - f^*) \Phi(z) + \sigma^{\text{asp}}(\alpha) \phi(z)]$, where $z = (\mu^{\text{asp}}(\alpha) - f^*) / \sigma^{\text{asp}}(\alpha)$.

Step 4 (Update): Evaluate the filtration for $\alpha_n^{\text{ex}_t}$ using the Lazy Filtration technique (edge-weight rescaling only, avoiding full Rips recomputation). Update the GP and repeat until convergence $|\alpha_n^{\text{ex}_t} - \alpha^n| < \varepsilon = 0.05$.

Output: $\alpha^* = \text{argmax}$ of the fitted GP posterior.

2.3.3 Connection to Theorem T3

Theorem T3 establishes that the topological discriminability ratio grows as $O(\alpha^2)$. The Bayesian optimization implicitly searches for the α regime where this $O(\alpha^2)$ amplification is maximized subject to the constraint that noise-induced variance remains bounded. In practice, the algorithm converges to $\alpha^* \in [1.5, 4.0]$ across the dynamical regimes tested.

2.3.4 Computational Cost

Full Rips recomputation for each α candidate would cost $O(n^3)$ per step. The Lazy Filtration technique — which rescales edge weights by a factor α^2 rather than recomputing the simplicial complex — reduces this to $O(n^2)$ per update. Combined with a typical convergence in 8–12 iterations, the total α -tuning overhead is approximately $2\times$ the cost of a single fixed- α filtration, making the approach computationally practical.

3. Persistent Homology, Persistent Entropy, and TAS

3.1 Vietoris-Rips Filtration and Barcodes

Using the metric space (P, d_α) , we construct the Vietoris-Rips filtration: for each radius $r \geq 0$, the simplicial complex $VR(P, r)$ includes all simplices with diameter $\leq r$. As r increases, topological features (connected components H_0 , loops H_1 , voids H_2) appear and disappear. Each feature is represented by a barcode $[birth_i, death_i)$. The persistence lifetime is:

$$L_i = death_i - birth_i; \quad L_{max} = \max_i L_i.$$

Long-lived features (large L_i) represent genuine topological structure. Short-lived features are topological noise. In the periodic regime, H_0 features are few and persistent (trajectory loops). In the chaotic regime, many transient features appear and disappear rapidly.

3.2 Persistent Entropy H_p

The normalized persistence distribution and entropy are:

$$p_i = L_i / \sum_j L_j, \quad H_p = -\sum_i p_i \cdot \log(p_i).$$

Interpretation: H_p is small when one feature dominates (simple topology, periodic regime). H_p is large when many features have comparable lifetimes (complex topology, chaotic regime). As the system transitions toward chaos, H_p begins rising before the Lyapunov exponent crosses its threshold — this is the core empirical finding of the paper.

The connection to our prior work [C]: H_p is the topological analogue of the fragmentation index $F(\epsilon)$, measuring the entropy of the near-optimal landscape structure.

3.3 Topological Anomaly Score (TAS)

TAS measures the emergence of unusually isolated topological structures:

$$TAS = (L_{max} - \mu_l) / \sigma_l,$$

where μ_l and σ_l are the mean and standard deviation of all barcode lifetimes. TAS is a standardized score: values near 0 indicate a homogeneous barcode distribution; large TAS indicates one outlier feature far from the baseline.

Revised interpretation (this version): Numerical experiments (Section 5) reveal that TAS crosses threshold at $t_{TAS} = 9.8s$, substantially later than both H_p (0.8s) and LLE (2.6s). This indicates that TAS is not an early-warning signal for initial chaos onset; rather, TAS detects sustained structural anomalies — isolated topological features that persist once the chaotic attractor is fully developed. This is analogous to the energy barrier H^* in [A]: H^* governs long-term mixing rather than short-term onset. Accordingly, TAS is best deployed as a structural stability monitor, while H_p serves as the early-warning indicator.

4. Theoretical Results

4.1 Jacobian Spectral Analysis

The double pendulum dynamics $\dot{X} = F(X)$ induces the Jacobian $J(X) = dF/dX$. Let $\lambda_{max}(J)$ denote its largest real eigenvalue. The trajectory separation satisfies:

$$|\delta X(t)| \leq C \cdot \exp\left[\int_0^t \lambda_{\max}(J(s)) ds\right] \cdot |\delta X(0)|.$$

Positive $\lambda_{\max}$ implies local stretching, which drives nearby trajectories apart and creates topological complexity in the phase-space point cloud.

Theorem T1 — Jacobian-Topology Bound [Proof Sketch; Rigorous Proof in Progress]

There exists a constant $C > 0$ depending on the initial condition radius r_0 such that:

$$L_{\max}(t) \leq C \cdot \exp\left[\int_0^t \lambda_{\max}(J(s)) ds\right].$$

Consequently: $\lambda_{\max} \uparrow \Rightarrow L_{\max} \uparrow \Rightarrow \text{TAS} \uparrow$.

Proof sketch: The maximum barcode lifetime bounds the maximum distance between connected components of P . By the trajectory separation inequality and the definition of $L_{\max}$ under $d\alpha$, the result follows from the Gronwall lemma applied to the distance between phase-space points.

Status: The above constitutes a heuristic argument (conjecture level). A complete proof requires (i) a rigorous connection between the Jacobian spectral norm and the Vietoris-Rips filtration diameter via the stability theorem of persistent homology, and (ii) quantitative bounds on the Gronwall constant C in terms of the system parameters. This proof is in preparation.

Connection to [A]: This theorem is the continuous-dynamics analogue of the Eyring-Kramers bound $\lambda_1(\beta) \sim \exp(-\beta \cdot H^*)$: just as H^* bounds the spectral gap, the integrated Jacobian bounds topological complexity growth.

4.2 Theorem T2: Corrected Statement

The original version of Theorem T2 stated the ordering $t_{\text{PE}} < t_{\text{TAS}} < t_{\text{LLE}}$. However, numerical experiments yield $t_{\text{PE}} = 0.8\text{s}$, $t_{\text{LLE}} = 2.6\text{s}$, $t_{\text{TAS}} = 9.8\text{s}$, giving the actual ordering $t_{\text{PE}} < t_{\text{LLE}} < t_{\text{TAS}}$. The original theorem was therefore inconsistent with the experimental results. We now state the corrected and empirically validated form:

Theorem T2 (Revised) — Persistent Entropy as Verified Early-Warning Signal

Under generic non-degeneracy conditions on the double pendulum trajectory, there exists a precursor interval $[t_0, t^c)$ such that $d/dt H_p > 0$ while the Largest Lyapunov Exponent (LLE) satisfies $\text{LLE} < \text{threshold}$. Therefore:

$$t_p^T < t_{\text{lle}},$$

where t_p^T and t_{lle} denote the respective threshold-crossing times of H_p and LLE.

Regarding TAS: $t_{\text{TAS}} > t_{\text{LLE}}$ in the current experiment. This is not a contradiction but a clarification of roles — TAS is a sustained-anomaly detector, not an initial-onset indicator. The revised claim is:

$$t_p^T < t_{\text{lle}} < t_{\text{TAS}} \quad (\text{sustained anomaly regime}).$$

Numerical validation: $t_p^T = 0.8\text{s}$, $t_{lle} = 2.6\text{s}$ (lead time: 1.8s). See Section 5.3.

Proof status: The inequality $t_p^T < t_{lle}$ is empirically confirmed across the chaotic initial condition tested. Analytical proof under non-degeneracy conditions requires establishing that topological entropy of P grows before the largest Lyapunov exponent becomes positive — a direction consistent with existing results on entropy as a chaos precursor (Gidea and Katz, 2018). Extension to multiple initial conditions is an open problem (see Section 8).

5. Numerical Experiments

5.1 Setup

Three dynamical regimes are investigated based on initial conditions: (1) Periodic: $y_0 = (0.5, 0, 0.5, 0)$; (2) Quasi-periodic: $y_0 = (1.0, 0, 1.0, 0)$; (3) Chaotic: $y_0 = (2.5, 0, 2.5, 0)$. Integration: RK45, $T = 40\text{s}$ (sliding-window analysis: $T = 60\text{s}$), $dt = 0.02\text{s}$, $rtol = 10^{-9}$. Anisotropic metric: $\alpha = 2.0$ (fixed; adaptive $\alpha^* = 1.92$ under the algorithm of Section 2.3). Sliding window: width = 80 states, stride = 30 states. Vietoris-Rips filtration on subsampled point clouds (every 2nd point).

Limitation: The experiments use three initial conditions. This constitutes a proof-of-concept rather than a statistically conclusive validation. Section 8 specifies the extended validation programme required for journal submission.

5.2 Regime Comparison

Regime	H_p (early)	H_p (late)	TAS (early)	TAS (late)	LLE
Periodic	8.521	8.522	2.650	2.902	0.046
Quasi-periodic	8.490	8.493	1.916	2.338	0.104
Chaotic	8.517	8.531	2.706	1.942	4.023

Table 1. Topological indicators across three regimes. H_p and TAS computed in early ($t < 8\text{s}$) and late ($t > 20\text{s}$) windows. Note: H_p differences between regimes are subtle; the early-warning signal is captured by the temporal trajectory (Figure 2), not the regime mean.

5.3 Sliding-Window Early Warning: Corrected Analysis

For the chaotic regime, H_p , TAS, and LLE are computed in a sliding window across the full trajectory ($T = 60\text{s}$). Threshold crossing times are defined as the first time the normalized indicator exceeds the 50th percentile of its time series.

Indicator	Threshold Crossing Time	Role	Lead vs LLE
Persistent Entropy H_p	$t_p^T = 0.8\text{ s}$	Early-warning signal	+ 1.8 s before LLE
LLE estimate	$t_{lle} = 2.6\text{ s}$	Reference (onset)	(baseline)

TAS	$t_{\text{TAS}} = 9.8 \text{ s}$	Sustained anomaly detector	$- 7.2 \text{ s (lags LLE)}$
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Table 2. Threshold crossing times for chaotic regime sliding-window analysis. The revised Theorem T2 reflects these results accurately.

Key finding: Persistent Entropy H_p provides 1.8 seconds of early warning before the LLE threshold crossing. This confirms the empirical part of Theorem T2 (revised). TAS lagging behind LLE is not a failure — it is a feature: TAS quantifies the emergence of large isolated topological structures characteristic of the fully-developed chaotic attractor, not the initial sensitivity growth.

5.4 Figures

Figure 1: Phase portraits (θ_1 vs ω_1) for three regimes. Periodic trajectory forms a closed curve; chaotic trajectory fills a complex region.

Figure 2: Sliding-window early warning comparison (normalized H_p , TAS, LLE vs time). H_p crosses threshold at $t = 0.8s$ — visible in the figure as a rising shoulder preceding the LLE step.

Figure 3: Bar comparison of topological indicators across three regimes.

6. Connection to Prior Work: The TLD Framework

This paper is part of the Topological Langevin Dynamics (TLD) research programme, which applies topological and energy barrier methods across continuous, arithmetic, and discrete domains. Table 3 summarizes the correspondence.

Concept	[A] Polynomial	[C] Combinatorial	Present Work
Energy function	$E(z) = P(z) ^2$	$E(x) = f(x)^2$	Phase-space cloud ($P, d\alpha$)
Complexity measure	H^* (saddle height)	$F(\epsilon)$ (fragmentation)	TAS, H_p
Main bound	$\lambda_1 \sim \exp(-\beta H^*)$	$t_{\text{mix}} \sim \exp(\beta H^*)$	$L_{\text{max}} \leq C \cdot \exp(\int \lambda_{\text{max}})$
Phase transition	$\beta^c = 1/H^*$	α^c (SAT)	t^c (chaos onset)
Early warning	No	Frag-RT corr $r = -0.894$	$t_p^T < t_{\text{lle}}$ by 1.8s
Status	Theorem (rigorous)	Conjecture + experiment	T1 sketch; T2 empirical; T3 sketch

Table 3. TLD framework correspondence across three papers.

7. Applications

The TAS- H_p framework applies to high-stakes domains where early chaos or regime-transition detection is critical.

Power Systems: Grid oscillations precede blackouts. TAS applied to AMI time-series point clouds could provide earlier warning than standard ADMS indicators — directly applicable to KEPCO distribution operations. The adaptive α -tuning algorithm (Section 2.3) enables domain-specific calibration without

manual parameter setting.

Climate Tipping Points: H_p computed on reanalysis trajectory clouds could identify precursors to abrupt regime transitions in ocean-atmosphere models, providing early-warning windows consistent with the 1.8s lead time demonstrated here (scaled to appropriate time units).

Aerospace Control: Flutter onset in flexible structures. The anisotropic metric can be tuned to velocity-sensitive modal coordinates — precisely the problem that α -tuning was designed for.

Daecheong Dam Algae Bloom: The authors' prior work on TAS-based algae prediction applies the same H_p framework to reservoir ecological dynamics.

Neuroscience: Epileptic seizure prediction. Topological changes in neural firing-rate clouds may precede the seizure onset — a direct analogue of the $t_p^T < t_{lle}$ relationship.

8. Limitations and Open Problems

The following limitations are acknowledged explicitly in response to reviewer recommendations. Each identifies a concrete path toward resolution.

L1 — Statistical Scope: Three Initial Conditions Only

All experiments use three initial conditions, one per dynamical regime. This is a proof-of-concept level validation. A reviewer will correctly ask whether the ordering $t_p^T < t_{lle}$ holds across 100 or 1000 initial conditions, and whether it is robust to observation noise.

Planned resolution: (a) Monte Carlo sampling of 500 initial conditions from the chaotic energy shell; (b) bootstrap confidence intervals on the lead time $\Delta t = t_{lle} - t_p^T$; (c) additive Gaussian noise test at SNR levels 10, 5, and 2 dB; (d) cross-system validation on the Lorenz, Rössler, and Duffing attractors.

L2 — Theorem T1 is a Proof Sketch (Conjecture Level)

Theorem T1 connects the integrated Jacobian norm to the maximum barcode lifetime via a Gronwall argument. This sketch does not constitute a rigorous proof. The missing steps are: (i) a quantitative stability bound connecting point-cloud diameter to barcode lifetimes (via the interleaving distance theorem of persistent homology), and (ii) a uniform constant C independent of the trajectory length T .

Planned resolution: Formal proof using the Stability Theorem of Edelsbrunner-Harer (2010) combined with the Gronwall lemma in the setting of the anisotropic Rips filtration. Expected for the next revision.

L3 — TAS Lags LLE in the Chaotic Experiment

TAS threshold crossing at $t_{TAS} = 9.8s$ lags behind LLE at $t_{LLE} = 2.6s$. This was incorrectly formulated in the original Theorem T2 as an early-warning claim. The corrected interpretation is that TAS measures sustained structural anomaly, not initial chaos onset.

Planned resolution: Explore a rolling-baseline TAS (using a local μ_i , σ_i estimated from the past k windows rather than the global baseline) which may respond earlier. Also explore H_1 (loop) barcodes as an additional early-warning channel.

O1 — Extension to H_1 and H_2 Barcodes

The present paper uses H_0 (connected components) barcodes. Full H_1 (loops) analysis may detect the emergence of chaotic tori before fragmentation. This is a natural next step within the TLD framework.

O2 — GPU Implementation for Real-Time Monitoring

Current implementation processes a 60s trajectory in ~ 20 s (CPU). Real-time monitoring of power systems or aerospace requires sub-second computation. GPU-accelerated Vietoris-Rips filtration (Ripser++ or Gudhi CUDA) is the primary technical bottleneck. The adaptive α algorithm is already formulated to minimize filtration recomputations via Lazy Filtration.

9. Conclusion

We proposed and experimentally validated a topology-based early-warning framework for chaos transition detection in double pendulum dynamics, combining Persistent Homology, Persistent Entropy H_p , and the Topological Anomaly Score TAS under an anisotropic metric with an adaptive tuning algorithm.

The central empirical finding stands: Persistent Entropy H_p provides 1.8 seconds of early warning before the Lyapunov exponent threshold crossing in the chaotic regime. This is confirmed by numerical sliding-window analysis and is consistent with Theorem T2 (revised).

Two important corrections are made relative to the original submission. First, Theorem T2 is corrected: the empirically validated ordering is $t_p^T < t_{le}$, with TAS acting as a sustained structural anomaly detector ($t_{TAS} > t_{le}$). This distinction is not a failure of TAS but a clarification of its role. Second, Theorem T1 is reframed as a proof sketch pending rigorous completion, rather than being presented as a complete theorem.

A new contribution in this revision is the adaptive α -tuning algorithm (Section 2.3), which uses Bayesian optimization with a Lazy Filtration update to automatically calibrate the anisotropic metric weight. This transforms the framework from a research prototype requiring manual parameter tuning into a self-calibrating system suitable for deployment in diverse dynamical environments.

The road to international journal publication requires: (1) rigorous proof of T1; (2) statistical validation across at least 100 initial conditions and multiple chaotic systems (Lorenz, Rössler, Duffing); (3) rolling-baseline TAS exploration; (4) GPU implementation for real-time use cases. These are well-defined research tasks, and the theoretical and experimental foundation established here provides a strong basis for their execution.

This work is part of the Topological Langevin Dynamics (TLD) research programme connecting energy barrier theory [A], arithmetic obstruction [B], and combinatorial fragmentation [C] into a unified framework. The consistent principle: energy barriers, topological fragmentation, and structural anomaly scores govern complexity — whether in optimization landscapes, integer solution spaces, or chaotic phase-

space dynamics.

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